

PULKIT CHATWAL

AI / Machine Learning Engineer

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AI Engineer bridging research and production, nine peer-reviewed publications in financial safety, multilingual NLP, and multimodal alignment, and part of founding team of [Vox Agent](#), a voice-agent deployed across enterprise clients fine-tuning and serving open-weight LLMs into systems that ship.

EDUCATION

Integrated Dual Degree (B.Tech + M.Tech), CSE & Artificial Intelligence

2021 - 2026

Rajiv Gandhi Institute of Petroleum Technology, Jaipur

PUBLICATIONS

Beyond Refusal: Measuring Safety Erosion in Multi-Turn Financial Jailbreaks

Under Review — EMNLP 2026

Pulkit Chatwal — Advised by Dr. Amit Agarwal (Capital One), Dr. Amit Kumar Jaiswal (Faculty Chair, IIT BHU), Mustafa Saify (Capital One)

- Built FinRed, a benchmark of 708 multi-turn adversarial conversations across five high-stakes financial domains (insider trading, money laundering, tax evasion, market manipulation, fiduciary breach) using five discourse strategies.
- Proposed Partial Compliance Rate (PCR) and Safety Erosion Rate (SER) to measure how early hedging in LLM responses erodes into later vulnerability across multi-turn financial jailbreaks.

Federated Multi-Task Holistic Affect Recognition via Bidirectional Cross-Modal Attention

Under Review — IEEE SLT 2026

Priyanshu, Pulkit Chatwal, Priyanshu Omkar Saraswat

- Proposed a federated multi-task framework fusing frozen Wav2Vec2 and RoBERTa embeddings via bidirectional cross-modal attention to jointly predict sentiment, emotion, humour, toxicity, and sarcasm without centralizing raw audio or transcripts.
- Showed multimodal fusion outperforms the best single modality by nearly 10 points ($p < 0.05$) across IEMOCAP, MuSETHS, and MELD under a seeded, significance-tested protocol.

Exploring Capability Thresholds in Ultra-Lightweight LLM Judges for Nugget-Based Report Evaluation

Jul 2026

Mann Bajpai, Pulkit Chatwal, Priyanshu Deswal, Harish Pratap Singh, Santosh Kumar Mishra — RAG4Reports Workshop, ACL 2026

- Built a zero-shot nugget-verification system for evaluating retrieval-grounded reports entirely on a single NVIDIA T4 GPU, comparing Qwen2-0.5B, Qwen2-1.5B, and Qwen2.5-0.5B as ultra-lightweight LLM judges.
- Ranked 13th of 21 teams; Qwen2.5-0.5B achieved $\tau_{gap} = 0.0772$ and Pearson $r = 0.2209$, showing model generation matters more than parameter count within this family.

Enhancing Causal Relationship Detection Using Prompt Engineering and LLMs

Jan 2025

Pulkit Chatwal, Amit Agarwal, Ankush Mittal — Joint Workshop of 9th FinNLP / 6th FNP / 1st LLMFinLegal, Abu Dhabi

- Applied zero-shot, few-shot, and chain-of-thought prompting on LLaMA 3.2 for financial causality extraction; achieved high semantic alignment across English and Spanish in the FinCausal 2025 shared task.

Overcoming Code-Mixing and Script-Mixing in Indian Language Summarization with Transformer Models

Dec 2024

Pulkit Chatwal, Amit Agarwal, Ankush Mittal — Working Notes of FIRE 2024, CEUR-WS.org

- Built multilingual summarization pipelines with fine-tuned mT5, IndicBART, and BART; ranked top in five languages and second in Telugu at FIRE 2024.

Cultura-Arabica: Probing and Enhancing Arabic Cultural Awareness in LLMs via LoRA

Nov 2025

Pulkit Chatwal, Santosh Kumar Mishra — Third Arabic NLP Conference: Shared Tasks

- Applied LoRA parameter-efficient fine-tuning and culturally informed prompt templates; achieved 71.65% accuracy and second position in PalmX 2025.

Meta Prompting for Analyst Report Generation: Turning Earnings Calls into Investment Guidance

Nov 2025

P. Chatwal, M. Bajpai, P. Deswal, H. P. Singh, S. K. Mishra — 10th Workshop on Financial Technology and NLP

- Proposed Structured Adversarial Synthesis (SAS), a multi-agent framework using Bull, Bear, and Devil's Advocate agents; ranked 2nd in Earnings2Insights FinNLP-2025 at EMNLP 2025.

From Theory to Action: The Role of an ESG-focused App in Transforming Sustainability Education

Jun 2024

P. Chatwal, A. Bakshi, S. Kumar, N. Agrawal, D. Som, N. Bhalla, A. Yadav — 15th ICCNT 2024, IEEE

- Designed and deployed a Flutter + Firebase ESG application with real-time sustainability indicators and gamified engagement; demonstrated improved adoption post-deployment.

RESEARCH PROJECTS

Safeguard: Country-Aware Legal Issues in LLMs

Ongoing

Advised by Dr. Yuxia Wang, Assistant Professor, INSAIT, Bulgaria

- Evaluating and adapting large language models under jurisdiction-specific AI legal constraints across eight countries: India, China, Egypt, Saudi Arabia, Nigeria, Bulgaria, and Bangladesh.

SKive-vLLM: Optimizing vLLM Serving via Paged Eviction and Flash Attention

Ongoing

Advised by Dr. Amit Agarwal (Capital One) and Dr. Amit Kumar Jaiswal (Faculty Chair, IIT BHU)

- Proposing SKive, a paged KV-cache eviction strategy combined with Flash Attention to optimize memory efficiency and throughput in vLLM-based inference serving.

FinRed: Unmasking Implicit Financial Toxicity and Regulatory Evasion in LLMs

Accepted

Advised by Dr. Amit Agarwal (Capital One India) — ACM TheWebConf 2026 Workshop (DHOWMiLLA)

- Built FinRed, a regulatory-evasion benchmark covering eight high-risk financial scenarios (e.g., insider trading, market manipulation).
- Evaluated Zero-Shot vs In-Context Learning across open-source LLMs; showed safety-aware ICL significantly improves refusal rates and regulatory compliance.

SKILLS

LLMs & GenAI: Prompt Engineering, LoRA, QLoRA, DPO, Instruct Tuning, RLHF, Quantization (AWQ/GPTQ)

RAG & Retrieval: RAG, Hybrid RAG, Graph RAG, Retrieval-Augmented Generation Pipelines

Model Serving & Inference: FastAPI, vLLM, TGI, llama.cpp, Triton Inference Server, SGLang, AirLLM

Speech & Voice: Speech-to-Text (STT), Text-to-Speech (TTS), Voice Agents

Agent Frameworks: LangGraph, LangChain, AutoGen, CrewAI

ML / DL: Deep Learning, Text Mining, Classification, Data Visualization, Federated Learning

Models Worked With: Claude, LLaMA, Qwen, OpenAI GPT, Gemini, Mistral, DeepSeek, mT5, IndicBART, BART, Wav2Vec2, RoBERTa....

Programming & APIs: Python, Git, VS Code, Jupyter, REST APIs, Third-Party Integration, Webhooks, OAuth

Libraries: PyTorch, Transformers, Scikit-learn, TensorFlow, NumPy, Pandas, BeautifulSoup

Cloud & DevOps: AWS, Azure, Docker

Databases & Automation: Vector Databases, PostgreSQL, Supabase, SQL, N8N, Zapier, Make

INDUSTRY EXPERIENCE

AI Engineer — Srinergy.ai

May 2025 – Present

Remote, Columbia

- Founding member and lead developer of [Voxx Agent](#), an AI voice-agent product currently deployed across multiple HyperKidz locations and Fidelity Direct Mortgage for real-time customer interaction and automated call handling.
- Integrated Voxx Agent with Roller, Acumatica, Optima Route, Salesforce, and Google Sheets to sync client workflows across different ERPs and CRMs.
- Designed and executed an outbound calling campaign for the Anup Jalota event in the Bay Area on behalf of the Shankara Eye Foundation, completing 9,000+ calls with optimized conversational flows and lead-engagement pipelines.

ACHIEVEMENTS

Conference Presenter — 4 International Conferences

2024 – 2025

- Presented accepted publications and research contributions at four international conferences/workshops: EMNLP (FinNLP), FIRE, the Arabic NLP Conference, and IEEE ICCCNT.

Smart India Hackathon Winner (2024) — Ministry of Education, Govt. of India

₹1,00,000

- Led the Mini Juris 2.0 team to build the ESG Post Office Initiative for digital sustainability transformation of Indian Postal Services — carbon tracking, QR-based smart energy control, AI solar-potential analysis, and the DakSathi chatbot.

Yukti Innovation Challenge Finalist (2024) — Ministry of Education, Govt. of India

National

- Selected nationally based on the ESG platform developed during Smart India Hackathon 2023.

Smart India Hackathon Joint Winner (2023) — Ministry of Education, Govt. of India

₹50,000

- Led an ESG implementation platform for higher-education institutions (Govt. of Jharkhand problem statement), resulting in a peer-reviewed IEEE publication (ICCCNT 2024).